WBSNet: Wavelet-Guided Boundary-Aware Skip Connections for Medical Image Segmentation

Research Proposal & Architecture Document

Prepared for: Guide Discussion & Hardware Planning

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1 Executive Summary

This document presents **WBSNet** (Wavelet Boundary Skip Network), a novel medical image segmentation architecture that addresses two fundamental weaknesses of U-Net's skip connections: the *semantic gap* between encoder and decoder features, and *boundary information loss* during downsampling. Our core contribution is the **WBS Module** (Wavelet Boundary Skip), a plug-and-play module that replaces standard skip connections by decomposing encoder features into frequency sub-bands using the Haar wavelet transform, applying targeted attention to each sub-band, and reconstructing enhanced features for the decoder.

The project targets publication at a medical image analysis venue (e.g., IEEE ISBI, MICCAI, or IEEE TMI) and requires GPU-based training infrastructure. This document provides the complete technical design, literature positioning, experiment plan, and hardware requirements for the research.

2 Problem Statement & Motivation

2.1 Why Medical Image Segmentation Matters

Medical image segmentation is the task of assigning a class label to every pixel in a medical image — for example, delineating a polyp in a colonoscopy frame or marking the boundary of a skin lesion in a dermoscopy image. Accurate segmentation is critical for:

- **Clinical diagnosis:** Precise boundary delineation affects staging (e.g., lesion size in skin cancer), treatment planning, and surgical navigation.
- **Automated screening:** Large-scale population screening (e.g., colorectal polyp detection) requires reliable pixel-level predictions.
- **Quantitative analysis:** Downstream measurements (volume, shape descriptors) depend on segmentation accuracy, especially at boundaries.

2.2 U-Net and the Skip Connection Problem

U-Net [Ronneberger et al., 2015] is the de facto standard for medical image segmentation. Its encoder-decoder architecture uses *skip connections* to transfer spatial information from the encoder to the decoder. However, two well-documented problems exist:

1. **Semantic gap:** Encoder features at shallow layers capture low-level patterns (edges, textures), while the decoder expects high-level semantic representations. Directly concatenating these creates a feature distribution mismatch that the decoder's limited capacity cannot fully resolve [Wang et al., 2024, Peng et al., 2023].
2. **Boundary information loss:** Progressive downsampling in the encoder erodes fine-grained boundary details. Standard skip connections pass raw encoder features without distinguishing boundary-critical information from noise, leading to blurred predictions at object edges [Bui et al., 2024, Authors, 2024a].

Key insight: Recent analytical work [Authors, 2024b] has shown that shallow skip connections often carry domain-specific noise that hurts cross-domain generalization. This motivates a *selective filtering* approach rather than passing all features blindly.

2.3 Why Existing Solutions Fall Short

Table 1 summarizes the landscape and identifies the specific gap our work fills.

Table 1: Gap analysis: How WBSNet differs from the closest related works.

Method	Wavelet in Skips?	Boundary Supervision?	Unified Module?	All Skip Levels?
U-Net v2 [Peng et al., 2023]	✗	✗	–	✓
UDTransNet [Wang et al., 2024]	✗	✗	✓	✓
WA-NET [Authors, 2025c]	Separate	Separate	✗	Partial
HCViT-Net [Authors, 2025a]	✓	✗	Partial	Only 1 level
MEGANet [Bui et al., 2024]	✗	Laplacian	✓	✓
WBSNet (Ours)	✓	✓	✓	✓

The gap: To the best of our knowledge, no existing method unifies wavelet-based frequency decomposition with boundary-supervised spatial attention *inside* the skip pathways at *all four* encoder-decoder fusion levels of a standard ResNet-U-Net topology.

3 Proposed Method: WBSNet

3.1 Architecture Overview

WBSNet follows a standard ResNet-U-Net encoder-decoder paradigm with a ResNet-34 [He et al., 2016] encoder pretrained on ImageNet. The key modification is that the **four skip pathways used by the decoder** are all replaced with our proposed WBS modules. Concretely, we use:

$$\{S_1, S_2, S_3, S_4\} = \{\text{stem/relu}, \text{layer1}, \text{layer2}, \text{layer3}\}, \quad B = \text{layer4} \tag{1}$$

$$\text{logits}_{\text{seg}} = \text{Head}\Big(\text{Up}_{2\times}\big(\text{Decoder}(B, \text{WBS}(S_1), \text{WBS}(S_2), \text{WBS}(S_3), \text{WBS}(S_4))\big)\Big) \tag{2}$$

where S_k denotes a skip-source feature and B is the deepest bottleneck representation. Importantly, **layer4 is used only as the bottleneck input**, not as an additional skip feature.

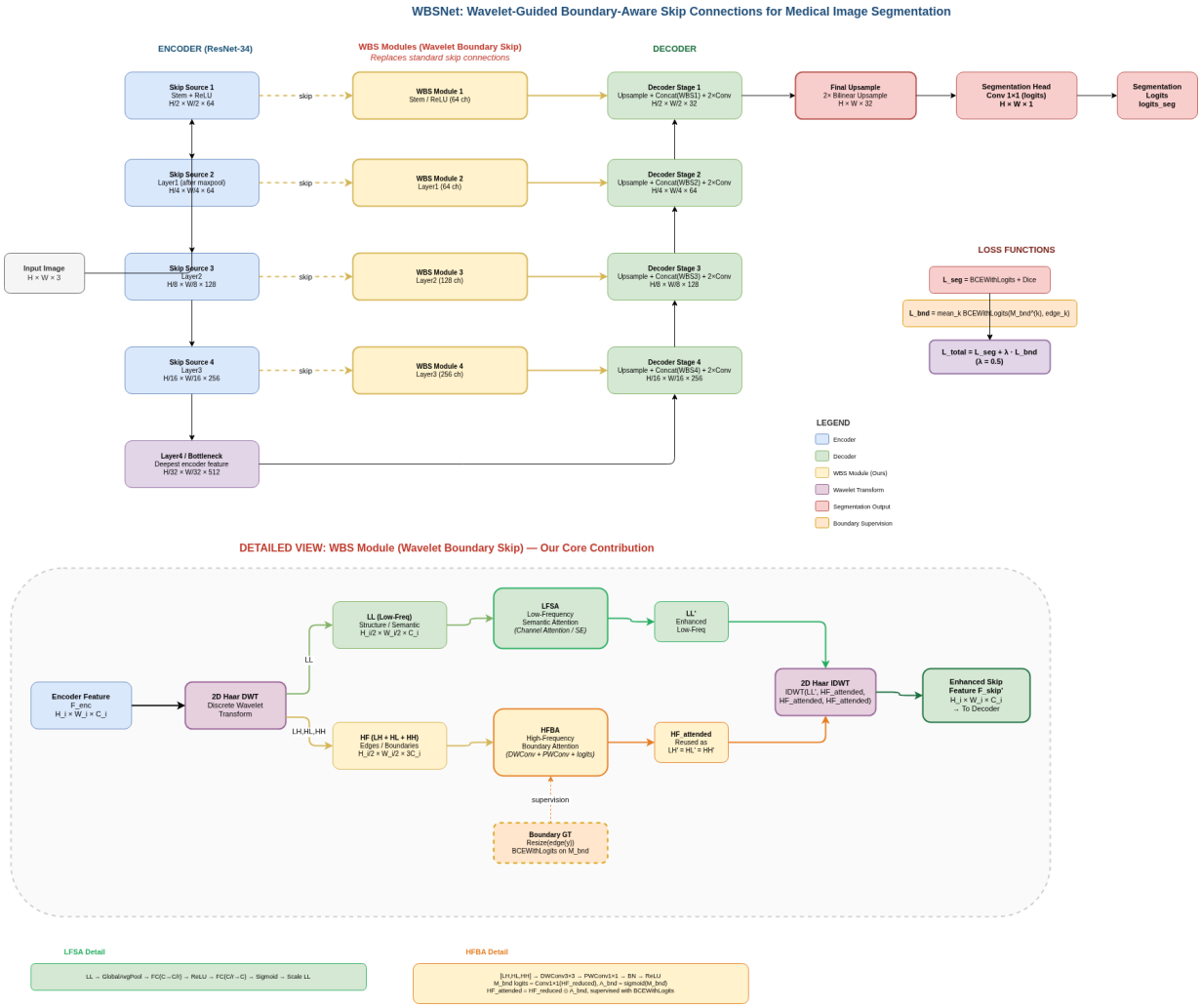

Figure 1: Overall architecture of WBSNet. Encoder features pass through WBS modules (yellow) which decompose features via DWT, apply LFSA and HFBA attention, and reconstruct via IDWT before forwarding to the decoder.

3.1.1 Encoder

We tap the standard ResNet-34 feature hierarchy at the following spatial scales:

- $S_1 = \text{stem/relu} \in \mathbb{R}^{B\times 64\times 176\times 176}$
- $S_2 = \text{layer1} \in \mathbb{R}^{B\times 64\times 88\times 88}$
- $S_3 = \text{layer2} \in \mathbb{R}^{B\times 128\times 44\times 44}$
- $S_4 = \text{layer3} \in \mathbb{R}^{B\times 256\times 22\times 22}$
- $B = \text{layer4} \in \mathbb{R}^{B\times 512\times 11\times 11}$

Thus, the decoder receives four WBS-refined skip tensors at resolutions 176, 88, 44, and 22, while the deepest 11×11 feature is reserved for the bottleneck.

3.1.2 Bottleneck

The output of `layer4` serves as the bottleneck representation $B \in \mathbb{R}^{B\times 512\times 11\times 11}$. No skip connection is taken from this stage.

3.1.3 Decoder

The decoder contains four stages:

- **Dec4:** $11 \to 22$, concatenate with $\mathrm{WBS}(S_4)$, output $22 \times 22 \times 256$
- **Dec3:** $22 \to 44$, concatenate with $\mathrm{WBS}(S_3)$, output $44 \times 44 \times 128$
- **Dec2:** $44 \to 88$, concatenate with $\mathrm{WBS}(S_2)$, output $88 \times 88 \times 64$
- **Dec1:** $88 \to 176$, concatenate with $\mathrm{WBS}(S_1)$, output $176 \times 176 \times 32$

After Dec1, a final $2\times$ bilinear upsampling produces a $352 \times 352 \times 32$ feature map before the segmentation head.

3.1.4 Output Heads

- **Segmentation head:** final $2\times$ upsampling $\to$ Conv 1×1, producing $\mathrm{logits}_{\mathrm{seg}} \in \mathbb{R}^{H\times W}$.

All WBS inputs in the final design have even spatial dimensions (176, 88, 44, 22), so no DWT padding is required.

3.2 The WBS Module — Core Contribution

Each WBS module takes an encoder feature $\mathbf{F}_{\mathrm{enc}} \in \mathbb{R}^{B\times C\times H\times W}$ and outputs an enhanced feature $\mathbf{F}'_{\mathrm{skip}} \in \mathbb{R}^{B\times C\times H\times W}$ of the *same shape*. The pipeline consists of four steps.

3.2.1 Step 1: 2D Haar Discrete Wavelet Transform (DWT)

The Haar DWT decomposes each channel of $\mathbf{F}_{\mathrm{enc}}$ into four frequency sub-bands:

$$\mathrm{LL}[i,j] = \tfrac{1}{2}\big(x[2i,2j] + x[2i,2j{+}1] + x[2i{+}1,2j] + x[2i{+}1,2j{+}1]\big) \tag{3}$$
$$\mathrm{LH}[i,j] = \tfrac{1}{2}\big(x[2i,2j] - x[2i,2j{+}1] + x[2i{+}1,2j] - x[2i{+}1,2j{+}1]\big) \tag{4}$$
$$\mathrm{HL}[i,j] = \tfrac{1}{2}\big(x[2i,2j] + x[2i,2j{+}1] - x[2i{+}1,2j] - x[2i{+}1,2j{+}1]\big) \tag{5}$$
$$\mathrm{HH}[i,j] = \tfrac{1}{2}\big(x[2i,2j] - x[2i,2j{+}1] - x[2i{+}1,2j] + x[2i{+}1,2j{+}1]\big) \tag{6}$$

- $\mathrm{LL} \in \mathbb{R}^{B\times C\times \frac{H}{2}\times \frac{W}{2}}$: Low-frequency approximation capturing *smooth structure and semantics*.
- $\mathrm{LH}, \mathrm{HL}, \mathrm{HH} \in \mathbb{R}^{B\times C\times \frac{H}{2}\times \frac{W}{2}}$: High-frequency details capturing *horizontal, vertical, and diagonal edges* respectively.

Why Haar? The Haar wavelet is computationally free (only additions and subtractions), perfectly invertible, and naturally aligns with boundary detection. It introduces **zero learnable parameters**.

3.2.2 Step 2: LFSA — Low-Frequency Semantic Attention

LFSA applies channel-wise recalibration (Squeeze-and-Excitation style [Hu et al., 2018]) to the LL sub-band, enhancing semantically important channels:

$$\mathrm{LL}' = \mathrm{LL} \odot \sigma\Big(\mathbf{W}_2 \cdot \mathrm{ReLU}\big(\mathbf{W}_1 \cdot \mathrm{GAP}(\mathrm{LL})\big)\Big) \tag{7}$$

where $\mathrm{GAP}(\cdot)$ is global average pooling, $\mathbf{W}_1 \in \mathbb{R}^{C/r \times C}$ and $\mathbf{W}_2 \in \mathbb{R}^{C \times C/r}$ are fully connected layers with reduction ratio $r = 16$, σ is the sigmoid function, and $\odot$ denotes channel-wise multiplication.

Intuition: Not all channels in the low-frequency sub-band are equally informative. LFSA learns to amplify channels carrying strong semantic signals and suppress noisy ones.

3.2.3 Step 3: HFBA — High-Frequency Boundary Attention

HFBA generates a spatial attention map that highlights boundary-relevant regions in the high-frequency sub-bands:

$$\mathbf{HF} = \mathrm{Concat}(\mathrm{LH}, \mathrm{HL}, \mathrm{HH}) \in \mathbb{R}^{B \times 3C \times \frac{H}{2} \times \frac{W}{2}} \tag{8}$$

$$\tilde{\mathbf{HF}} = \mathrm{ReLU}\Big(\mathrm{BN}\big(\mathrm{PWConv}_{1\times1}(\mathrm{DWConv}_{3\times3}(\mathbf{HF}))\big)\Big) \in \mathbb{R}^{B \times C \times \frac{H}{2} \times \frac{W}{2}} \tag{9}$$

$$\mathbf{M}_{\mathrm{bnd}}^{(k)} = \mathrm{Conv}_{1\times1}(\tilde{\mathbf{HF}}) \in \mathbb{R}^{B \times 1 \times \frac{H}{2} \times \frac{W}{2}} \tag{10}$$

$$\mathbf{A}_{\mathrm{bnd}}^{(k)} = \sigma\big(\mathbf{M}_{\mathrm{bnd}}^{(k)}\big) \tag{11}$$

$$\mathbf{HF}' = \tilde{\mathbf{HF}} \odot \mathbf{A}_{\mathrm{bnd}}^{(k)} \tag{12}$$

where DWConv and PWConv denote depthwise and pointwise separable convolutions respectively (lightweight variant).

Critical design: $\mathbf{M}_{\mathrm{bnd}}^{(k)}$ denotes the *logits* produced by the k-th WBS module. A sigmoid is applied only to obtain the attention weights $\mathbf{A}_{\mathrm{bnd}}^{(k)}$ used for feature modulation. During training, the logits are supervised by boundary ground truth derived from Sobel edge detection on the segmentation mask:

$$\mathcal{L}_{\mathrm{bnd}}^{(k)} = \mathrm{BCEWithLogitsLoss}\big(\mathbf{M}_{\mathrm{bnd}}^{(k)},\ \mathrm{Resize}(\mathrm{Sobel}(\mathbf{y}), \mathrm{size}(\mathbf{M}_{\mathrm{bnd}}^{(k)}))\big) \tag{13}$$

This explicit boundary supervision forces the attention map to focus on actual anatomical boundaries rather than arbitrary high-frequency noise.

3.2.4 Step 4: Inverse DWT (IDWT)

The enhanced sub-bands are recombined via the inverse Haar wavelet transform:

$$\mathbf{F}'_{\mathrm{skip}} = \mathrm{IDWT}(\mathrm{LL}', \mathrm{LH}', \mathrm{HL}', \mathrm{HH}') \in \mathbb{R}^{B \times C \times H \times W} \tag{14}$$

Following the lightweight design, the same attended tensor is reused for all three high-frequency slots:

$$\mathrm{LH}' = \mathrm{HL}' = \mathrm{HH}' = \mathbf{HF}'.$$

This produces a feature map at the *original spatial resolution*, ready for concatenation with the upsampled decoder feature.

3.3 Loss Function

The total training loss combines segmentation quality and boundary fidelity:

$$\mathcal{L}_{\text{seg}} = \text{BCEWithLogitsLoss}(\text{logits}_{\text{seg}}, y) + \text{Dice}\big(\sigma(\text{logits}_{\text{seg}}), y\big) \tag{15}$$

$$\mathcal{L}_{\text{bnd}} = \frac{1}{4}\sum_{k=1}^{4} \text{BCEWithLogitsLoss}\Big(\mathbf{M}_{\text{bnd}}^{(k)}, \text{Resize}(\text{Sobel}(y), \text{size}(\mathbf{M}_{\text{bnd}}^{(k)}))\Big) \tag{16}$$

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{seg}} + \lambda \cdot \mathcal{L}_{\text{bnd}} \tag{17}$$

where $\lambda = 0.5$ balances the two objectives (tuned during ablation).

4 Novelty & Contributions

Our primary contributions are:

1. **WBS Module:** A novel, plug-and-play skip connection enhancement module that unifies wavelet-based frequency decomposition with boundary-supervised spatial attention. To the best of our knowledge, this is the first lightweight design to do so across all four skip pathways of a standard ResNet-U-Net.

2. **Dual-path attention in frequency domain:** LFSA for semantic channel recalibration of the low-frequency sub-band, and HFBA for boundary-aware spatial attention on the high-frequency sub-bands — each path addresses a distinct aspect of the semantic gap problem.

3. **Boundary GT supervision of attention maps:** Unlike prior methods that learn attention in an unsupervised manner, we directly supervise the high-frequency attention logits with Sobel-derived boundary ground truth, providing an explicit learning signal for boundary awareness.

4. **Comprehensive evaluation:** Experiments on polyp segmentation (Kvasir-SEG, CVC-ClinicDB, CVC-ColonDB) and skin lesion segmentation (ISIC 2018) benchmarks, with ablation studies isolating each component's contribution.

5 Experiment Design

5.1 Datasets

Table 2: Datasets for evaluation.

Dataset	Domain	Modality	Train	Test	Resolution
Kvasir-SEG	Gastrointestinal polyp	Colonoscopy	880	120	352×352
CVC-ClinicDB	Gastrointestinal polyp	Colonoscopy	550	62	352×352
CVC-ColonDB	Gastrointestinal polyp	Colonoscopy	–	380	352×352
ISIC 2018	Skin lesion	Dermoscopy	2594	1000	352×352

Cross-dataset generalization: We train on Kvasir-SEG and evaluate zero-shot on CVC-ColonDB to test whether the WBS module improves generalization (not just in-distribution performance).

5.2 Baselines

Table 3: Baseline methods for comparison (all using ResNet-34 backbone for fair comparison).

Method	Category	Why Included
U-Net [Ronneberger et al., 2015]	Standard baseline	Foundation architecture
U-Net++ [Zhou et al., 2018]	Dense skip connections	Skip connection baseline
Attention U-Net [Oktay et al., 2018]	Attention gates	Attention-based skip filtering
PraNet [Fan et al., 2020]	Polyp-specific	Strong domain-specific model
U-Net v2 [Peng et al., 2023]	Enhanced skip connections	Direct skip connection competitor
TransUNet [Chen et al., 2021]	Transformer encoder	Transformer baseline
MEGANet [Bui et al., 2024]	Edge-guided attention	Edge-aware competitor

5.3 Evaluation Metrics

Table 4: Evaluation metrics.

Metric	Formula	What It Measures
Dice (DSC)	$\frac{2 \cdot TP}{2 \cdot TP + FP + FN}$	Overall segmentation overlap
IoU (Jaccard)	$\frac{TP}{TP + FP + FN}$	Intersection over union
Precision	$\frac{TP}{TP + FP}$	False positive rate
Recall	$\frac{TP}{TP + FN}$	False negative rate
HD95	95th %-ile Hausdorff	**Boundary accuracy** (critical for our claims)

5.4 Ablation Study

Table 5: Ablation variants to isolate each contribution.

ID	Variant	What It Tests
A1	U-Net + Standard Skip	Baseline (no WBS module)
A2	U-Net + WBS (Full)	Complete WBSNet
A3	WBS without HFBA	Is boundary attention needed?
A4	WBS without LFSA	Is channel attention on LL needed?
A5	WBS without boundary supervision	Is explicit boundary GT supervision needed?
A6	Attention on raw skips (no wavelet)	**Is wavelet decomposition itself needed?**
A7	WBS with Daubechies-2 wavelet	Does wavelet family matter?

A6 is the most critical ablation. If removing the wavelet decomposition shows similar performance, the core novelty claim is weakened. This ablation proves that *operating in the frequency domain* is fundamentally different from just adding more attention to raw features.

5.5 Training Configuration

Table 6: Training hyperparameters.

Parameter	Value
Optimizer	AdamW
Learning rate (encoder / decoder+WBS)	1×10^{-4} / 1×10^{-3}
LR scheduler	CosineAnnealing ($T_{\max}$ = epochs)
Batch size	16
Epochs	200 (polyp datasets), 100 (ISIC 2018)
Image size	352×352
Augmentation	HFlip, VFlip, Rotate90, ColorJitter, RandomResizedCrop
Boundary loss weight λ	0.5
SE reduction ratio r	16
Encoder pretrained	ImageNet (ResNet-34)

6 Parameter & Computational Analysis

6.1 Parameter Overhead of WBS Modules

The WBS module adds parameters only through the LFSA and HFBA sub-modules (DWT/IDWT are parameter-free). Using the recommended **depthwise separable** HFBA variant:

Table 7: Parameter overhead per WBS module (depthwise separable HFBA).

Stage	C	LFSA Params	HFBA Params	Total
WBS 1	64	512	∼13K	∼13.5K
WBS 2	128	2K	∼50K	∼52K
WBS 3	256	8K	∼200K	∼208K
WBS 4	512	33K	∼790K	∼823K
			Total additional parameters	**∼1.1M**

Context: ResNet-34 has ∼21.8M parameters. The WBS modules add ∼5% overhead, which is negligible compared to transformer-based alternatives (e.g., UDTransNet adds ∼8M for its DAT modules).

6.2 Memory & FLOPs Estimation

- **DWT doubles feature maps temporarily**: At each WBS module, the DWT produces 4 sub-bands at half resolution. Net memory per module: ∼2× the encoder feature at that stage.
- **Peak GPU memory** (batch size 16, 352 × 352 input, ResNet-34 encoder): Estimated **∼8–10 GB** based on similar architectures.
- **Training throughput**: Expected ∼15–25 images/second on a single GPU (comparable to standard U-Net with attention gates).

7 Hardware Requirements

This section provides detailed hardware specifications needed to train WBSNet, covering minimum requirements, recommended setup, and estimated training times.

7.1 GPU Requirements

Table 8: GPU requirements for training WBSNet.

Specification	Minimum	Recommended	Ideal
GPU Model	NVIDIA RTX 3060 (12GB)	NVIDIA RTX 3090 / A5000 (24GB)	NVIDIA A100 (40/80GB)
VRAM	12 GB	24 GB	40+ GB
Batch Size	4–8	16	32+
Mixed Precision	Required (FP16)	Recommended	Optional
Multi-GPU	Not needed	Optional (faster)	2–4× for ablations

7.2 Complete System Requirements

Table 9: Full system specifications.

Component	Minimum	Recommended
GPU	1× RTX 3060 (12GB)	1× RTX 3090/4090 (24GB)
CPU	8-core (Intel i7 / AMD Ryzen 7)	16-core
RAM	16 GB	32 GB
Storage	100 GB SSD	500 GB NVMe SSD
CUDA Version	$\geq$ 11.7	$\geq$ 12.0
PyTorch	$\geq$ 2.0	$\geq$ 2.1
Python	$\geq$ 3.8	$\geq$ 3.10

7.3 Training Time Estimates

Table 10: Estimated training time per dataset (single run, single GPU).

Dataset	Epochs	RTX 3060 (12GB)	RTX 3090 (24GB)
Kvasir-SEG (880 train)	200	∼4–5 hours	∼2–3 hours
CVC-ClinicDB (550 train)	200	∼3–4 hours	∼1.5–2 hours
ISIC 2018 (2594 train)	100	∼6–8 hours	∼3–4 hours
All main experiments (4 datasets, 1 run each)		∼15–20 hours	∼8–10 hours
Ablation studies (7 variants × 1 dataset)		∼28–35 hours	∼14–21 hours
Statistical runs (3 runs per experiment)		×3 multiplier	×3 multiplier
TOTAL ESTIMATED GPU TIME		**∼130–165 hours**	**∼66–93 hours**

7.4 Cloud GPU Alternatives

If local GPUs are unavailable, the following cloud options are viable:

Table 11: Cloud GPU pricing estimates (approximate, subject to change).

Platform	GPU	Cost/Hour (USD)	Est. Total Cost
Google Colab Pro+	A100 (40GB)	∼$0.50–1.00	$50–100
Kaggle (free tier)	T4/P100 (16GB)	Free (30h/week)	Free (slow)
Lambda Cloud	A100 (80GB)	∼$1.10	$75–110
Vast.ai	RTX 3090	∼$0.20–0.40	$15–40
RunPod	RTX 4090	∼$0.40–0.70	$30–70

7.5 Software Dependencies

Table 12: Key Python packages.

Package	Version	Purpose
`torch`	≥ 2.0	Deep learning framework
`torchvision`	≥ 0.15	Pretrained encoders
`segmentation-models-pytorch`	$\geq 0.3.3$	Baseline models
`pytorch-wavelets`	≥ 1.3	DWT/IDWT implementation
`albumentations`	≥ 1.3	Data augmentation
`opencv-python`	≥ 4.8	Sobel edge, image processing
`scipy`	≥ 1.10	HD95 metric computation
`wandb` or `tensorboard`	latest	Experiment tracking

8 Related Work Landscape

Our work sits at the intersection of three research directions:

8.1 Skip Connection Design

The semantic gap in U-Net skip connections has been addressed by multiple approaches: U-Net++ [Zhou et al., 2018] adds dense nested skip pathways, U-Net v2 [Peng et al., 2023] uses Hadamard product-based semantic infusion, UDTransNet [Wang et al., 2024] employs dual-attention transformers, and SK-VM++ [Wu et al., 2025] uses Mamba-based refinement. However, none of these operate in the frequency domain or explicitly target boundary information.

8.2 Wavelet / Frequency Methods in Segmentation

Wavelet transforms have been used in medical image analysis for feature extraction [Authors, 2025b], noise reduction [Authors, 2023c], and feature fusion [Li et al., 2024]. SFFNet [Authors, 2024c] combines Haar wavelet decomposition with dual-cross attention for remote sensing. WaveFormer [Authors, 2023b] reformulates self-attention in the frequency domain. However, to the best of our knowledge, **no prior work applies wavelet decomposition specifically inside all four skip pathways of a standard ResNet-U-Net** with boundary supervision.

8.3 Boundary-Aware Segmentation

Boundary enhancement has been explored via Laplacian operators (MEGANet [Bui et al., 2024]), dedicated boundary branches (BRNet [Authors, 2024a], BMANet [Zhu et al., 2025]), uncertainty modeling (BUNet [Authors, 2023a]), and wavelet-based auxiliary modules (WA-NET [Authors, 2025c], SkinMamba [Zhu et al., 2024]). The closest work, HCViT-Net [Authors, 2025a], uses wavelet attention in skip connections but only at a single resolution level and without boundary supervision.

Our differentiation: To the best of our knowledge, WBSNet is the first lightweight design to unify wavelet decomposition, dual-path attention, and boundary GT supervision in a single module applied at all four skip pathways of a standard ResNet-U-Net.

9 Project Timeline

Table 13: 7-week project timeline.

Week	Phase	Tasks
1	Setup & Baseline	Environment setup, download datasets, preprocess boundary GTs, train vanilla U-Net baseline
2	Core Implementation	Implement DWT/IDWT, LFSA, HFBA, complete WBS module, integrate into U-Net
3	Main Experiments	Train WBSNet on all 4 datasets, compare with all baselines
4	Ablation Studies	Run all 7 ablation variants, generate qualitative results
5	Paper Writing I	Draft Introduction, Related Work, Methodology sections
6	Paper Writing II	Draft Experiments, Conclusion, create all figures and tables
7	Polish & Submit	Proofreading, formatting, reference verification, submission

10 Expected Outcomes

Based on the performance of related methods in the literature, we expect:

- **Dice improvement:** +1–3% over vanilla U-Net (ResNet-34) and competitive with or exceeding U-Net v2, Attention U-Net, and MEGANet on polyp and skin lesion benchmarks.
- **HD95 improvement:** Significant reduction (lower = better) due to explicit boundary supervision — this is our strongest expected signal.
- **Cross-dataset generalization:** Improved zero-shot performance on CVC-ColonDB when trained on Kvasir-SEG, demonstrating that wavelet-filtered skip features are less prone to dataset-specific overfitting.
- **Lightweight overhead:** $\sim$1.1M additional parameters ($\sim$5% of ResNet-34), making the module practical for clinical deployment.

11 Key Risks & Mitigations

Table 14: Risk assessment.

Risk	Impact	Mitigation
Ablation A6 (no wavelet) shows similar performance	Core novelty weakened	Test multiple attention baselines; if wavelet doesn't help, pivot to stronger wavelet families
GPU memory overflow at batch 16	Slower training	Use mixed precision (FP16); reduce batch to 8 with gradient accumulation
Marginal Dice improvement	Paper less compelling	Focus on HD95 and qualitative boundary quality; target boundary-focused venues
Concurrent competing publication	Reduced novelty	Differentiate via comprehensive ablation and plug-and-play demonstration on multiple backbones

12 Code Structure

```
WBSNet/
 models/
    wbsnet.py          # Main model
    wbs_module.py      # WBS Module (DWT + LFSA + HFBA + IDWT)
    lfsa.py            # Low-Frequency Semantic Attention
    hfba.py            # High-Frequency Boundary Attention
    wavelet.py         # DWT and IDWT implementations
    decoder.py         # Decoder blocks
 datasets/
    polyp_dataset.py   # Kvasir-SEG, CVC-ClinicDB loader
    isic_dataset.py    # ISIC 2018 loader
    transforms.py      # Data augmentation
 losses/
    seg_loss.py        # BCEWithLogits + Dice
    boundary_loss.py   # Per-module boundary BCEWithLogits
 utils/
    metrics.py         # Dice, IoU, HD95 calculation
    boundary_gt.py     # Sobel edge GT generation
    visualize.py       # Qualitative result visualization
 train.py               # Training script
```

```
test.py                 # Evaluation script
configs/
    kvasir.yaml         # Dataset-specific config
    isic.yaml
```

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